



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: MCA

Semester: II

Name of the Paper: Advance Database Management System

Paper Code: TMC 201

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)

- a. Describe the three-schema architecture of database. How the different layers of the three-schema architecture related to the concept of data independence? CO1

OR

- b. If you are a database analyst for Oracle Corporation, as a stake holder of the corporation to train the customer on database environment, list the functionalities and components of database system. CO1

Q2. (10 Marks)

- a. Explain the different concepts of Extended ER diagram. Define the various constraints that can be specified while using generalization. CO1

OR

- b. Define the following terms:

- i. Cardinality Ratio
- ii. Participation constraint
- iii. Three codd's rule.
- iv. Domain Integrity Constraint
- v. Referential Integrity Constraint

CO1

Q3. (10 Marks)

- a. Draw an ER Diagram for the following Scenario

"An academic institution is affiliated to a university. The institution possesses several departments; each department offer several courses. Each department has its own infrastructure where several teachers teach several students".

- i. Transform your ER diagram into relational database.
- ii. Specify the rules that are applied to minimize the number of tables while transforming ER diagram to database. CO1

OR

- b. i. Discuss the role of database administrator. CO1

ii. As per the rules of the company, how can you limit the entry of such ambiguous data in to the table at the database level? Illustrate with Constraints for the following statements

- i. Every supplier should have a name.
- ii. Every supplier should have a supplier code that is distinct, and the length of the supplier code



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should be equal to six digits.

iii. Quantity of product supplied should be greater than zero and less than 200.

iv. Order date should be less than the current date.

Q4. (10 Marks)

a. Explain the different steps briefly that are involved in processing a high-level query. CO2

OR

b. Consider the relational algebra expression: CO2

$\sigma_{\text{Dept}=\text{"CSE"}}(\text{Employee} \bowtie \text{Department})$

(a) Explain how the query is evaluated.

(b) Apply query transformation rules to optimize it.

Q5. (10 Marks)

a. Write queries for the following in SQL using the database below: CO3

Employee (EID, Name, DeptID, Sal)

DEPARTMENT (DeptID, DeptName)

i. Create the above tables implementing foreign key relationship.

ii. List the Name and salary of employees having salary between 1000 and 5000

iii. Display the EID, Name, and DeptID of employees belonging to department 'CSE'.

iv. Show the Name of all employees along with their Department Name.

v. List the employees whose name do not include 'a'.

OR

b. What is normalization? Why it is required. Define 1NF, 2NF and 3NF with example. CO2